

Dual LiDAR-Based MAV Tracking System

Extension from Single-Sensor to Dual-Sensor Detection, Calibration, and Fusion

1. Motivation: Limitations of the Single-LiDAR System

The original drone detection system used a single Livox Mid-360 mounted inside a fixed indoor enclosure, applying background-subtraction based clustering to isolate and track a small micro aerial vehicle (MAV). While this configuration validated the core detection pipeline successfully, indoor drone tracking imposes stricter precision and coverage requirements than outdoor or open-field detection, and the single-sensor setup exposed three compounding limitations.

First, effective detection range was limited to approximately 3 metres from the sensor. Beyond this distance, the small physical cross-section of the MAV produced too few LiDAR returns per scan for reliable clustering, causing intermittent or total loss of track in the far half of the enclosure.

Second, the confined dimensions of the indoor test volume amplified the practical impact of this range limit: with a pit only a few metres across, a large fraction of the usable flight volume fell outside the sensor's reliable detection radius, leaving minimal safe margin for manoeuvring.

Third, precision requirements for indoor tracking are inherently tighter than for outdoor applications, since the tracked volume itself is small and any positional jitter represents a proportionally larger fraction of the available space. The single-sensor system exhibited increased frame-to-frame jitter at extended range, driven by the reduced point density on the MAV, undermining the precision the application demands.

2. Transition to a Dual-LiDAR Configuration

To resolve the range and jitter limitations, a second Livox Mid-360 was introduced, with the two sensors positioned in diagonally opposite corners of the enclosure. This configuration ensures that every point within the flight volume lies within reliable detection range of at least one sensor, directly addressing the range limitation identified in the single-sensor system, while the increased aggregate point density on the MAV in regions seen by both sensors further reduces jitter.

This extension, however, introduces a new structural limitation of its own: each Mid-360 has a vertical field of view that does not extend directly overhead, producing a conical blind region above each sensor. In a single-sensor configuration this blind cone represents a fixed, unavoidable gap in coverage. With two sensors positioned in opposite corners, each sensor's blind cone falls within the other sensor's field of view, so the dual configuration is specifically intended to eliminate this gap — provided the two sensors are accurately calibrated to a common reference frame, which subsequently proved to be the dominant engineering challenge of the dual-sensor system.

3. Dual-LiDAR Technical Setup

3.1 Removal of FAST-LIO

The single-sensor system relied on FAST-LIO2, a tightly-coupled LiDAR-inertial odometry framework, to perform motion undistortion of each scan and to register point clouds into a fixed world frame. FAST-LIO2 is architected around a single LiDAR paired with a single IMU and does not provide a mechanism to fuse two independent LiDAR-IMU streams into one consistent state estimate. Running two separate FAST-LIO2 instances was considered and rejected, since each instance would produce its own independently drifting odometry frame, requiring continuous re-alignment between the two frames and reintroducing exactly the kind of inter-sensor disagreement the system needed to eliminate.

Because both Mid-360 units in the dual configuration are rigidly mounted and static for the duration of operation, the motion-undistortion capability of FAST-LIO2 is unnecessary: undistortion exists to correct for sensor motion during a scan, and a stationary sensor has none to correct. FAST-LIO2 was therefore removed entirely from the dual-sensor pipeline, in favour of consuming each sensor's raw point cloud directly and applying a fixed, manually calibrated extrinsic transform per sensor into a shared reference frame. This simplifies the software stack, removes a source of odometry drift, and removes the single-IMU fusion limitation entirely.

3.2 Native Dual-Sensor Support in the Livox Driver

During implementation it was found that the `livox_ros_driver2` package already provides native support for operating multiple Mid-360 units from a single driver instance, removing the need for any custom message-splitting or conversion logic. With the driver's `multi_topic` launch option enabled, the driver automatically publishes a separate, independent `PointCloud2` topic per connected sensor, each named according to that sensor's IP address (for example, distinct topics for the units at 192.168.1.117 and 192.168.1.122). Combined with setting the driver's transfer format to standard `PointCloud2` rather than the Livox-specific `CustomMsg` format — which is only required when per-point timestamps are needed for motion undistortion, itself no longer necessary following the removal of FAST-LIO — this allowed both sensors to be served by one driver process with no additional translation layer required.

3.3 Network Configuration and Extrinsic Calibration in Livox Viewer

Each Mid-360 unit was assigned a distinct static IP address (192.168.1.117 and 192.168.1.122) and configured within Livox Viewer 2's Device Manager, allowing both units to be monitored and configured simultaneously from a single interface. Livox Viewer 2 additionally exposes an Extrinsic Parameter panel per connected LiDAR, allowing roll, pitch, yaw, and X/Y/Z translation to be entered directly and applied live, with the effect immediately visible in the combined point-cloud viewport.

This feature was used as a first-pass visualisation and sanity-check tool: with both sensors displayed simultaneously and colour-coded by LiDAR ID, coarse extrinsic values could be entered and adjusted interactively to bring the two point clouds into approximate visual agreement before any values were transferred into the ROS 2 pipeline. This provided an immediate, qualitative check — a shared wall or floor surface visibly split into two offset layers indicates the two sensors disagree on its position, while a single fused surface indicates approximate agreement — well before the more precise, quantitative calibration steps described in Section 6 were carried out.

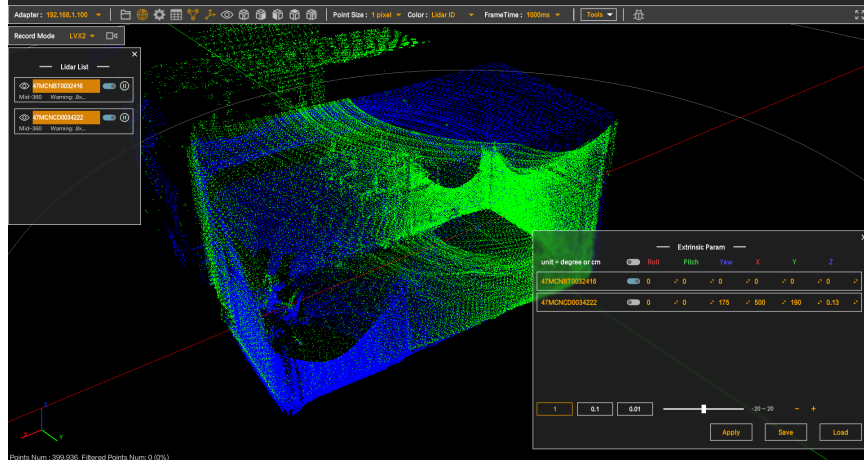


Fig. Dual lidar preliminary extrinsic calibration in LivoxViewer

4. Package Architecture: dual_lidar_tracker

The dual_lidar_tracker ROS 2 package extends the original single-sensor tracker to consume and fuse two independent point-cloud streams. It retains the previously established background-subtraction and clustering pipeline for isolating the MAV from its static surroundings — voxel-based background modelling, dynamic point isolation, frame accumulation, and Euclidean clustering — which is not repeated in this document, having been fully specified in the single-sensor methodology. This section addresses only what is new in the dual-sensor extension: the mathematics of bringing two independently posed sensors into a common frame, and the preliminary per-sensor detection that precedes fusion.

4.1 Common Reference Frame

A shared coordinate system, the pit frame, was defined with its origin at a fixed physical corner of the enclosure at floor level, with the X axis running along the long dimension of the pit and the Y axis along the short dimension. Each sensor's physical position and orientation within this frame is described by a pose vector:

$$P_i = (x_i, y_i, z_i, \psi_i), \quad i \in \{1, 2\}$$

where (x_i, y_i, z_i) is the translational offset of sensor i from the pit-frame origin and ψ_i is its yaw about the vertical axis. Sensor roll and pitch are assumed to be zero under the assumption that each unit is physically mounted level; this assumption and its practical consequences are revisited in Section 7.

4.2 Per-Sensor Point Transform

Each incoming point p_{lidar} , expressed in sensor i 's own local frame, is mapped into the shared pit frame by a rigid-body transform composed of a yaw rotation and a translation:

$$p_{pit} = R_i \cdot p_{lidar} + t_i$$

with the rotation matrix and translation vector defined as:

$$R_i = \begin{bmatrix} \cos \psi_i & -\sin \psi_i & 0 \\ \sin \psi_i & \cos \psi_i & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad t_i = (x_i, y_i, z_i)^T$$

This transform is applied independently to every point in each sensor's incoming cloud before any further processing, so that all subsequent stages of the pipeline — region-of-interest cropping, background subtraction, and clustering — operate entirely within the shared pit frame, regardless of which physical sensor a given point originated from.

4.3 Preliminary Per-Sensor Detection

Following transformation into the pit frame, each sensor's cloud is independently cropped to the region of interest, background-subtracted, and clustered, yielding at most one candidate MAV cluster C_i per sensor per frame, together with its centroid:

$$c_i = (1 / |C_i|) \cdot \sum p, \quad p \in C_i$$

This per-sensor centroid, together with the number of contributing points $|C_i|$, is the fundamental measurement unit that the fusion stage described in Section 6 operates on. Critically, this centroid is computed and made available independently for each sensor, prior to any merging step — the significance of this design choice, and the failure mode it was introduced to correct, is the subject of Section 6.

5. The Fused View Node

A dedicated visualisation node, the cloud fuser, was developed to provide a live, colour-coded merged view of both sensors' point clouds for calibration and inspection purposes, independent of the detection pipeline itself. This node subscribes directly to both raw driver topics, applies the same per-sensor extrinsic transform described in Section 4.2 to each incoming cloud, and republishes a single merged cloud in the pit frame, with every point additionally tagged with a colour according to its originating sensor — conventionally green for the first unit and orange for the second, mirroring the Lidar-ID colour convention used in Livox Viewer 2.

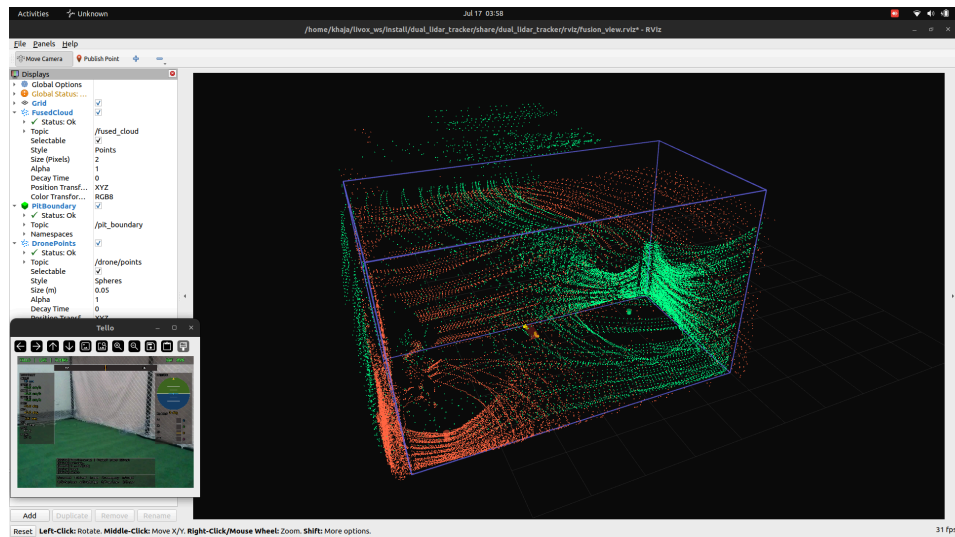


Fig. Fused view node and rviz visualization.

The essential purpose of this node is to make extrinsic misalignment visually diagnosable: if the two sensors' poses are correctly calibrated, any single static surface common to both fields of view — most usefully a wall or the floor — appears as one continuous, single- coloured-pair surface with green and orange points interleaved. If a residual extrinsic error exists, that same surface appears as two visibly separated, parallel sheets, one per colour, with the magnitude and direction of the separation directly indicating the size and nature of the pose error — a uniform offset indicating a translational error, and a separation that grows across the surface indicating a yaw error.

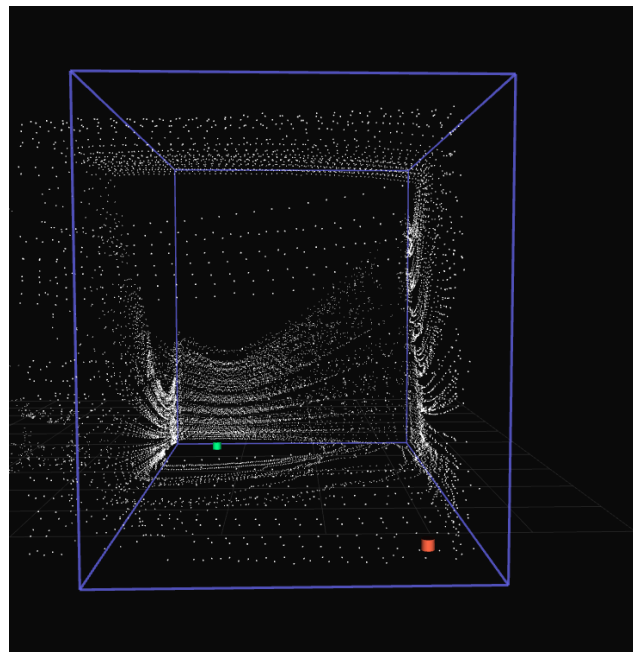


Fig. Uncalibrated misaligned point cloud view with respect to pit domain

A wireframe boundary representing the physical, independently tape-measured dimensions of the pit is published alongside the fused cloud, providing a fixed, ground-truth reference frame against which each sensor's calibration could be validated — not only against each other, but against the physical enclosure itself. This distinction proved important: two sensors can appear to agree with each other while both being offset identically from the true pit geometry, an error that a wall-referenced boundary box exposes but a purely inter-sensor comparison cannot. The extrinsic parameters used by the fuser are therefore not derived from comparing the two LiDARs to one another, but from independently calibrating each sensor against manually measured wall, floor, and corner positions of the pit, obtained by clicking reference points on each sensor's individual cloud in RViz2 and comparing them against the physically measured pit dimensions.

6. The Central Problem: Overlap-Zone Phasing

The dominant issue encountered in the dual-sensor system emerged specifically in the central region of the pit visible to both sensors simultaneously. While tracking was stable and accurate when the MAV was within either sensor's near-field domain, position estimates in the overlap zone exhibited severe

instability, characterised by abrupt jumps between two or more apparent positions rather than a smooth trajectory — a behaviour referred to throughout development as phasing.

6.1 Root-Cause Analysis

Investigation identified three compounding contributors to this failure mode.

- Point-count-weighted centroid bias: the original pipeline merged both sensors' point clouds prior to computing a single centroid. Because the Mid-360's non-repetitive scan pattern produces a randomly fluctuating number of returns on a small target from one frame to the next, the relative contribution of each sensor to the merged centroid varied unpredictably frame to frame, causing the estimated position to swing toward whichever sensor happened to dominate the point count in that instant.
- Residual extrinsic misalignment: small, uncorrected errors in each sensor's calibrated pose mean the two sensors' independent views of the MAV disagree by some fixed offset. This disagreement is invisible near either sensor individually, where one sensor's view dominates the merged cloud, but becomes the dominant error source precisely in the overlap zone, where both views contribute comparably.
- Temporal misalignment: clouds from the two sensors were merged within a bounded time window. For a moving target, this permitted comparatively stale data from one sensor to be fused with fresher data from the other, introducing an additional apparent displacement proportional to the MAV's velocity.

6.2 Fusion Approach: Per-Sensor Kalman Weighting

Rather than merging raw point clouds prior to measurement, the architecture was restructured so that each sensor's cloud is independently reduced to its own centroid, as described in Section 4.3, and each centroid is then submitted to the tracking Kalman filter as a separate, independently timestamped measurement, rather than as a single combined observation. This directly removes the temporal misalignment contributor, since each measurement is processed at its own true acquisition time rather than being held for a fixed merge window.

To address the point-count-weighting contributor, each per-sensor measurement is assigned a measurement noise inversely related to its supporting point count, so that low-confidence detections are automatically down-weighted by the filter rather than contributing equally regardless of reliability:

$$r_i = r_0 \cdot \sqrt{(\min(16, \max(1, n_{ref}/n_i)))}$$

where n_i is the number of points supporting sensor i 's detected cluster in the current frame, n_{ref} is a nominal reference point count, and r_0 is the filter's baseline measurement noise. This replaces the discrete, unstable point-ratio weighting inherent to cloud merging with a smooth, statistically principled equivalent, in which the Kalman filter's own update equations perform the weighting rather than the raw point ratio of a merged cloud.

6.3 Centroid Bias Calibration

The Kalman-weighted fusion approach reduces sensitivity to point-count fluctuation, but does not, by itself, correct a genuine residual extrinsic disagreement between the two sensors — such an offset must be measured directly and removed at its source. An online bias monitor was implemented for this purpose, computing a running exponential moving average of the vector difference between the two sensors' centroids whenever both detect the MAV within a shared time window:

$$b \leftarrow (1 - \alpha) \cdot b + \alpha \cdot (c_2 - c_1)$$

with α a small smoothing constant. A converged, stable value of b directly represents the residual translational disagreement between the two sensors' calibrated poses. Calibration was performed empirically by placing the MAV, ideally on a static stand for repeatability, at the centre of the overlap zone and allowing this average to converge over several hundred paired detections, then subtracting the converged bias directly from the second sensor's configured translational pose:

$$(x_2, y_2, z_2)_{corrected} = (x_2, y_2, z_2) - b$$

In practice, this procedure converged to a stable offset within several hundred sample pairs and, following correction, reduced inter-sensor disagreement from several centimetres to a residual within a few centimetres of the sensor's own baseline range noise — the remaining difference being consistent with the physical body depth of the MAV itself, since each sensor necessarily observes a different face of the same physical object from its respective corner.

7. Data Acquisition Frequency and Present Sources of Error

The present system operates at the native 10 Hz frame rate of the Mid-360 sensors, with each sensor contributing an independent measurement to the fused track at its own acquisition time, giving an effective combined update rate of up to 20 Hz within the overlap zone where both sensors detect simultaneously, tapering to the single-sensor 10 Hz rate elsewhere in the pit. Several sources of residual error and open refinement work remain, summarised below.

- Sub-centimetre fusion accuracy: the current extrinsic calibration reduces inter-sensor disagreement to within a few centimetres, adequate for coarse position and velocity tracking, but insufficient for archival-quality or millimetre-precision measurement applications. Achieving mm-to-cm level fusion accuracy will require a more precise calibration procedure than manual wall-clicking and centroid-bias averaging alone, potentially incorporating a dedicated calibration target and a least-squares extrinsic optimisation across many observations rather than a single converged bias vector.
- Roll and pitch correction: the current extrinsic model assumes each sensor is mounted perfectly level, with roll and pitch fixed at zero. Physical mounting is never perfectly level in practice, and an uncorrected tilt was observed to project a flat floor as a sloped surface in the fused view. Precise archival-grade fusion will require extending the extrinsic model to a full six-degree-of-freedom pose per sensor, with roll and pitch measured and corrected alongside yaw and translation.
- Fusion-induced jitter management: although per-sensor Kalman weighting substantially reduces the phasing behaviour described in Section 6, some residual jitter remains at the boundary of the

overlap zone, where the set of contributing sensors changes discretely from one to two. Smoothing this transition, for example through a continuous confidence-based blending rather than a hard inclusion/exclusion of a sensor's measurement, remains an area for refinement.

- Inaccuracy domain polishing: calibration accuracy was validated primarily at the pit centre and along the central overlap region. Systematic verification of positional accuracy across the full pit volume, including near the physical edges and corners furthest from either sensor, has not yet been completed, and may reveal additional position-dependent residual errors requiring further correction.
- Increasing acquisition frequency: the 10 Hz per-sensor rate is a hardware limitation of the Mid-360 itself and cannot be increased directly. Effective update rate could nonetheless be improved by reducing the frame-accumulation window used to compensate for sparse point returns on the MAV, which would lower measurement latency at the cost of point density per detection, or by introducing asynchronous, per-sensor Kalman prediction between measurements so that the filter's output update rate is decoupled from the sensors' native scan rate entirely.

8. Summary

The dual-LiDAR system was developed to overcome the range and precision limitations of the single-sensor configuration, in which effective MAV detection was confined to approximately three metres and degraded further at range due to the confined indoor test volume. Positioning two Livox Mid-360 units in opposite corners of the pit extended reliable coverage across the full flight volume and, in principle, allowed each sensor's vertical blind cone to be compensated by the other's field of view, at the cost of introducing a new engineering requirement: the two sensors had to be brought into accurate agreement within a single shared reference frame.

This was achieved by removing FAST-LIO2, which offered no meaningful benefit once both sensors were static, leveraging the Livox driver's native ability to serve two units as independent PointCloud2 topics without custom message handling, and calibrating each sensor's extrinsic pose independently against manually measured pit geometry rather than against the other sensor directly. A dedicated fused-view node made this calibration visually verifiable, rendering both sensors' clouds in a common, colour-coded pit frame against a ground-truth wireframe boundary.

The most significant obstacle encountered was phasing in the region visible to both sensors, traced to three compounding causes: point-count-driven centroid bias in the merged cloud, residual extrinsic disagreement between the sensors, and temporal misalignment introduced by merging clouds within a fixed time window. The resolution restructured detection around independent per-sensor centroids fused as separately timestamped, confidence-weighted Kalman measurements, combined with an online bias monitor used to measure and subtract the residual inter-sensor offset directly from the second sensor's configured pose. Together these changes reduced the overlap-zone instability to a level consistent with the sensors' own baseline noise.

The system as it stands operates reliably at the native 10 Hz per-sensor rate with centimetre-level inter-sensor agreement, sufficient for stable position and velocity tracking of the MAV throughout the

pit. Remaining work is concentrated on tightening this agreement toward millimetre-to-centimetre precision, extending the extrinsic model to account for roll and pitch rather than yaw and translation alone, smoothing the transition at the boundary of the overlap zone, validating accuracy across the full pit volume rather than primarily at its centre, and increasing the effective measurement rate through reduced frame accumulation and asynchronous filter prediction.

9. Data Analysis

No ground truth for this data collection, purely raw data with visual validation that the data is coherent and reasonable.

Data of three domains were taken:

1. Domain of Lidar 1
2. Domain of Lidar 2
3. Common domain / Middle domain

Here, by domain it is meant where the lidars detection range is highest and the density of the point cloud is optimal.

Data is not valid because they have no ground of comparison, thus not shown here. Updated data shall be taken with coherent rubric.